

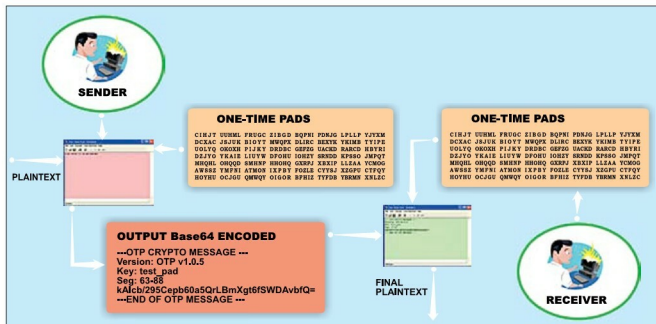


Fig. 1: One-time-pad crypto system

only if the key tape is unpredictable, as long as the message and used only once, the message would be secure. Moreover, the encryption proved to be unbreakable and hence the one-time pad was born.

Of all the methods of encryption ever devised, only Vernam cipher or one-time pad has been mathematically proved to be completely secure. The worth of all other ciphers is based on computational security. If a cipher is computationally secure, this means the probability of cracking the encryption key using current computational technology and algorithms within a reasonable time is supposedly extremely small, yet not impossible. In theory, every cryptographic algorithm, except for the one-time pad, can be broken, given enough ciphertext and time.

How to use one-time pads

One-time pads are used in pairs. The more the copies of a given pad, the greater is the likelihood that one may be captured, in which case the system is completely broken. One copy of the pad is kept by each user, and pads must be exchanged via a secure channel, such as during a face-to-face meeting. Pads must only be used once.

The fastest method of encrypting a

message with a one-time pad is with a computer. If you choose this method, keep the pad on a CD or DVD and destroy it completely once used. Supposedly deleted data can be retrieved and reconstructed from magnetic storage, so never store pads on your hard drive. The recipient of the message should take the same precautions. Never use a networked computer for implementing the encryption because of possible eavesdropping.

One-time pad follows four important rules:

1. The key is as long as the message or data that is encrypted.
2. The key is truly random (not generated by a simple computer function or such).
3. Each key is used only once, and both sender and receiver must destroy their key after use.
4. There should only be two copies of the key—one for the sender and the other for the receiver (some exceptions exist for multiple receivers).

If these rules are followed correctly, one-time pads can be proven unbreakable, according to Claude Shannon's 'Communication Theory of Secrecy Systems.'

If each key letter in the pad sequence is truly random, a cryptanalyst

can do no better than try every possible key letter for every ciphertext message position. This is a hopeless situation for the attacker because it is equivalent to trying all the possible messages the key could ever encrypt. Even for a short pad also, the ciphertext can provide no clue as to which one of the possibilities is the real message.

One-time pads or one-time encryption is not to be confused with one-time keys or one-time passwords. Such one-time keys, limited in size, are valid only for a single encryption session by some crypto-algorithm under control of that key. Small one-time keys are by no means unbreakable, because the security of the encryption depends on the crypto algorithm they are used for.

As long as the pads are unique and never reused, no statistical analysis or pattern matching techniques can be applied by cryptanalysts. The fact that the pad can be used only once is the 'one time' point of this cipher.

Soviet intelligence once reused one-time pads years after they had originally been distributed to field agents in Britain. The British intelligence service noticed some patterns in coded messages and began searching for comparisons through a complete archive of all encrypted communica-